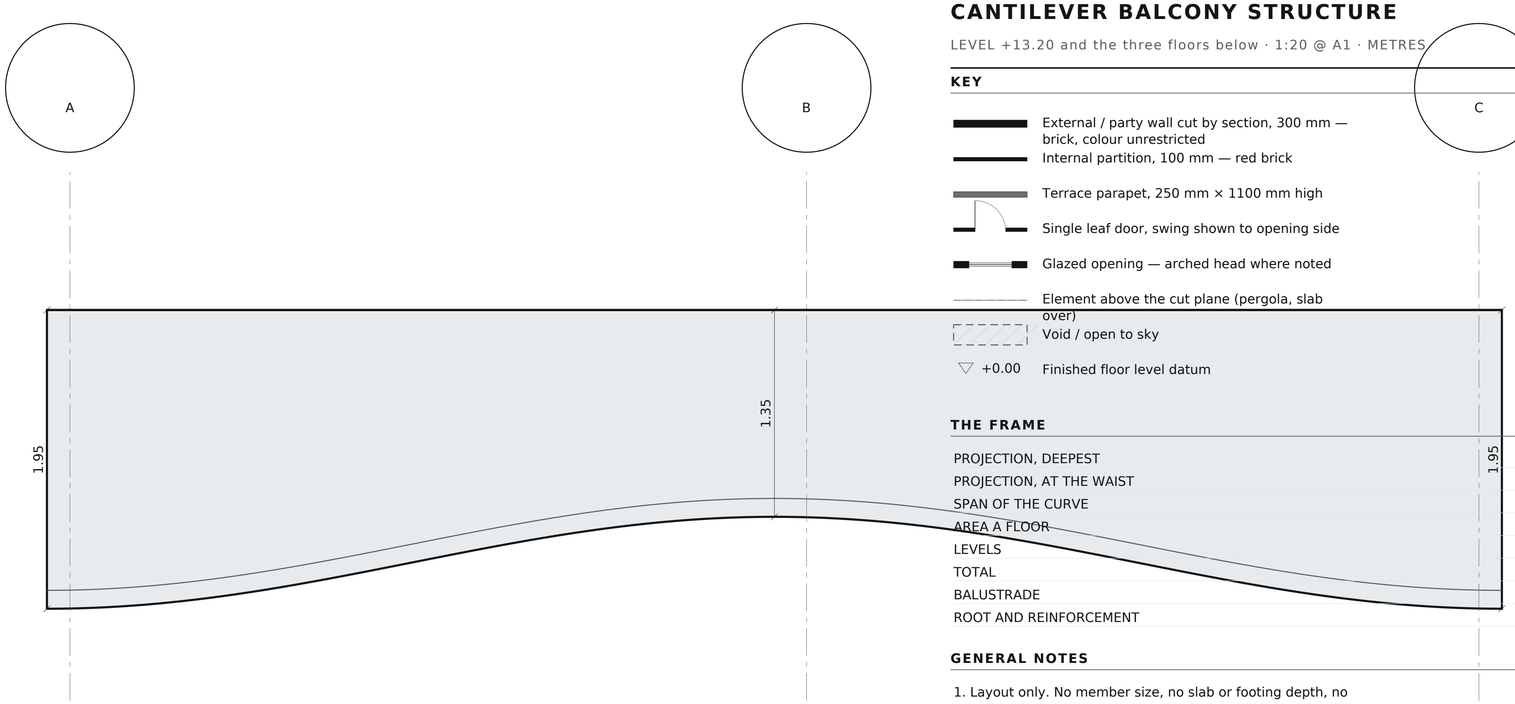
# CANTILEVER BALCONY STRUCTURE

LEVEL +13.20 and the three floors below · 1:20 @ A1 · METRES




SLAB EDGE — THICKNESS, ROOT DEPTH AND REINFORCEMENT BY THE ENGINEER

## KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- +0.00 Finished floor level datum

## THE FRAME

| | |
|---|---|
| PROJECTION, DEEPEST | 1.95 m |
| PROJECTION, AT THE WAIST | 1.35 m |
| SPAN OF THE CURVE | 9.50 m |
| AREA A FLOOR | 15.7 m² |
| LEVELS | 4 |
| TOTAL | 62.7 m² |
| BALUSTRADE | glass, 120 mm inboard |
| ROOT AND REINFORCEMENT | by the engineer |

## GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. A 1.95 m cantilever on a curve, repeated on four levels, is the single most engineering-sensitive element in this building. It is drawn at 1:20 for that reason and for no other.

4. The curve, the span, the projection at every point and the balustrade line are exact. The slab thickness, the root depth into the floor behind, the top steel and the deflection and vibration checks are entirely the engineer's.

5. Grid line 1 runs along the face the cantilever springs from. The balcony projects south of it and is unsupported at its far edge on all four levels.

**FIVE-STOREY VACATION VILLA** — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

**CANTILEVER BALCONY STRUCTURE**

| DRAWING No | REVISION | SCALE | DATE |
|---|---|---|---|
| **STR-500** | **P21** | **1:20 @ A1** | **2026-10-10** |

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor
P20 2026-10-07 internal partitions 100 mm red brick, specified by the client
P21 2026-10-07 300 mm external and party walls: brick, colour unrestricted